

MT8002 Elements of Set Theory  
The Zermelo–Fraenkel Axioms  
Axioms 7 to 9

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# Preliminaries

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## Textbook

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study.

## Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

# Outline

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Introduction

Inductive Sets

The Axiom of Infinite

The Construction of the Set of the Natural Numbers

The Axiom of Replacement

The Axiom of Foundation

References

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## Introduction

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# Introduction

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## Formal language

Recall that in ZFC the concepts of **pure set** and **membership relation** are primitive (undefined). Also recall that the **universe of discourse** of the theory is the **pure sets**.

# Introduction

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Recall that in ZFC the concepts of **pure set** and **membership relation** are primitive (undefined). Also recall that the **universe of discourse** of the theory is the **pure sets**.

## Classification

- Axioms stating the existence of sets
- Axioms determining properties of sets
- Axioms for building sets from other sets

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# Inductive Sets

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## Definition (p. 38)

Let  $x$  be a set. The **successor** of  $x$ , denoted  $x^+$ , is the set defined by

$$x^+ := x \cup \{x\}.$$



# Inductive Sets

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## Example

$$\begin{aligned}\emptyset^+ &\doteq \emptyset \cup \{\emptyset\} \\ &\doteq \{\emptyset\},\end{aligned}$$

$$\begin{aligned}\emptyset^{++} &\doteq (\emptyset^+)^+ \\ &\doteq \{\emptyset\}^+ \\ &\doteq \{\emptyset\} \cup \{\{\emptyset\}\} \\ &\doteq \{\emptyset, \{\emptyset\}\},\end{aligned}$$

$$\begin{aligned}\emptyset^{+++} &\doteq (\emptyset^{++})^+ \\ &\doteq \{\emptyset, \{\emptyset\}\}^+ \\ &\doteq \{\emptyset, \{\emptyset\}\} \cup \{\{\emptyset, \{\emptyset\}\}\} \\ &\doteq \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}.\end{aligned}$$

# Inductive Sets

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## Definition (p. 39)

A set  $y$  is **inductive** if

- (a)  $\emptyset \in y$ ,
- (b) if  $x \in y$  then  $x^+ \in y$ .

# Inductive Sets

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Or, equivalently,

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# Inductive Sets

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## Remark

Note that an inductive set is an infinite set.

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# The Axiom of Infinite

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## Axiom of Infinite

$$\exists x (\emptyset \in x \wedge \forall y (y \in x \rightarrow y \cup \{y\} \in x)).$$

# The Axiom of Infinite

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## Axiom of Infinite

$\exists x (\emptyset \in x \wedge \forall y (y \in x \rightarrow y \cup \{y\} \in x)).$

Informal description

*“There is an inductive set.” (p. 92)*

Category

Axiom stating the existence of set.

# The Axiom of Infinite

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## Example

Let  $x$  be an inductive set by the Axiom of Infinite. Then  $x^+, x^{++}, x^{+++}, \dots$  are also inductive sets.



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# The Construction of the Set of the Natural Numbers

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## Defining the natural numbers

For defining the natural numbers in ZFC we need to define zero and successor following the Dedekind–Peano axioms.

# The Construction of the Set of the Natural Numbers

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$$0 := \emptyset.$$

# The Construction of the Set of the Natural Numbers

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## Defining the natural numbers

For defining the natural numbers in ZFC we need to define zero and successor following the Dedekind–Peano axioms.

### Definition (p. 38)

$\mathbf{0} := \emptyset$ .

### Definition

A **numeral**  $\mathbf{n}$  is the representation of a natural number  $n$  in a formal system.

# The Construction of the Set of the Natural Numbers

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## Example

Some numerals from zero and successor.

$$\mathbf{1} := \mathbf{0}^+ \doteq \mathbf{0} \cup \{\mathbf{0}\} \quad \doteq \{\mathbf{0}\},$$

$$\mathbf{2} := \mathbf{1}^+ \doteq \mathbf{1} \cup \{\mathbf{1}\} \doteq \{\mathbf{0}\} \cup \{\mathbf{1}\} \quad \doteq \{\mathbf{0}, \mathbf{1}\},$$

$$\mathbf{3} := \mathbf{2}^+ \doteq \mathbf{2} \cup \{\mathbf{2}\} \doteq \{\mathbf{0}, \mathbf{1}\} \cup \{\mathbf{2}\} \doteq \{\mathbf{0}, \mathbf{1}, \mathbf{2}\}.$$

# The Construction of the Set of the Natural Numbers

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## Example

Some numerals from zero and successor.

$$\begin{aligned} \mathbf{1} &:= \mathbf{0}^+ \doteq \mathbf{0} \cup \{\mathbf{0}\} && \doteq \{\mathbf{0}\}, \\ \mathbf{2} &:= \mathbf{1}^+ \doteq \mathbf{1} \cup \{\mathbf{1}\} \doteq \{\mathbf{0}\} \cup \{\mathbf{1}\} && \doteq \{\mathbf{0}, \mathbf{1}\}, \\ \mathbf{3} &:= \mathbf{2}^+ \doteq \mathbf{2} \cup \{\mathbf{2}\} \doteq \{\mathbf{0}, \mathbf{1}\} \cup \{\mathbf{2}\} && \doteq \{\mathbf{0}, \mathbf{1}, \mathbf{2}\}. \end{aligned}$$

The same numerals from the empty set and successor.

$$\begin{aligned} \mathbf{1} &\doteq \emptyset^+ && \doteq \{\emptyset\}, \\ \mathbf{2} &\doteq \emptyset^{++} && \doteq \{\emptyset, \{\emptyset\}\}, \\ \mathbf{3} &\doteq \emptyset^{+++} && \doteq \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}. \end{aligned}$$

# The Construction of the Set of the Natural Numbers

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## Definition (p. 40)

Let  $y$  be any inductive set. The **set of the natural numbers**, denoted  $\omega$ , is the intersection of all inductive subsets of  $y$ , that is,

$$\omega := \bigcap \{ z : z \text{ is an inductive subset of } y \}, \quad (\text{naive definition}),$$

$$\omega := \bigcap \{ z \in \mathcal{P}(y) : z \text{ is inductive} \}, \quad (\text{ZFC definition}).$$

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## Definition (p. 40)

A **natural number** is a member of  $\omega$ .

## Notation

The textbook denotes the set of natural numbers by  $\omega$ . We use the standard set theory notation to emphasise the difference between the two sets.

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# The Axiom of Replacement

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## Axiom of Replacement

$$\forall x \exists y \forall y' (y' \in y \leftrightarrow \exists x' (x' \in x \wedge \phi(x', y'))),$$

where  $\phi(s, t)$  is a formula (possibly referring to named sets (possibly with parameters)) such that  $\forall s \exists t (\phi(s, t) \wedge \forall t' (\phi(s, t') \rightarrow t' \doteq t))$ .

# The Axiom of Replacement

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where  $\phi(s, t)$  is a formula (possibly referring to named sets (possibly with parameters)) such that  $\forall s \exists t (\phi(s, t) \wedge \forall t' (\phi(s, t') \rightarrow t' \doteq t))$ .

### Informal description

*“If  $\phi(s, t)$  is a **class** function, then when its domain is restricted to [a set]  $x$ , the resulting images form a set  $y$ .” (p. 92)*

### Category

Axiom for building a set from other set.

# The Axiom of Replacement

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## Example

In this example we show that from a formula (class function)  $\phi(s, t)$  we can define a proper class. See Exercise 4.39.

# The Axiom of Replacement

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For example, let  $\phi(s, t) : s \doteq t$  be a formula (class function). From  $\phi(s, t)$  we can define the proper class

$$\mathcal{A} := \{ \langle s, s \rangle : s \text{ is a set and } \phi(s, s) \text{ holds} \}.$$

# The Axiom of Replacement

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## Example

In this example we show that from a formula (class function)  $\phi(s, t)$  we can define a proper class. See Exercise 4.39.

For example, let  $\phi(s, t) : s \doteq t$  be a formula (class function). From  $\phi(s, t)$  we can define the proper class

$$\mathcal{A} := \{ \langle s, s \rangle : s \text{ is a set and } \phi(s, s) \text{ holds} \}.$$

Note that  $\mathcal{A}$  is a proper class because the universe of sets  $\mathcal{V}$  is the proper class of first coordinates of the ordered pairs it contains.

# The Axiom of Replacement

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## Exercise 4.23

Let  $x$  be a set. Show that  $\{ \{y\} : y \in x \}$  is a set.



# The Axiom of Replacement

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## Exercise 4.23

Let  $x$  be a set. Show that  $\{ \{y\} : y \in x \}$  is a set.

## Solution

Let  $\phi(s, t) : t \doteq \{s\}$  be a formula (class function).

By the Axiom of Replacement when the formula (class function)  $\phi(s, t)$  is restricted to the set  $x$  there is a set formed with the images of the class function, that is,  $\{ \{y\} : y \in x \}$  is a set.

# The Axiom of Replacement

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## Exercise

In this exercise we show that we can define an infinite set using the Axiom of Replacement.

# The Axiom of Replacement

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## Exercise

In this exercise we show that we can define an infinite set using the Axiom of Replacement.

The first levels of the cumulative hierarchy of sets are defined by

$$\begin{aligned}V_0 &:= \emptyset, \\ V_{n+1} &:= \mathcal{P}(V_n), \text{ for } n \in \omega.\end{aligned}$$

Show that  $\{ V_n : n \in \omega \}$  is a set.

(continued on next slide)

# The Axiom of Replacement

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## Solution

We define a formula (class function) by

$$\phi(n, x) : \exists f (f \text{ is a function} \wedge \text{Dom}(f) \doteq n^+ \wedge f(\mathbf{0}) \doteq \emptyset \wedge \\ \forall k (k \in n \rightarrow f(k^+) \doteq \mathcal{P}(f(k))) \wedge f(n) \doteq x).$$

By the Axiom of Replacement when restricting  $\phi(n, x)$  to the set  $\omega$  we get the set

$$A := \{ x : \exists \mathbf{n} (\mathbf{n} \in \omega \wedge \phi(\mathbf{n}, x)) \},$$

which is the required set, that is,

$$\{ V_{\mathbf{n}} : \mathbf{n} \in \omega \} \doteq A.$$

# The Axiom of Replacement

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## Technical remark

Since there is an instance of the Axiom of Replacement for each formula (class function)  $\phi(s, t)$  such that  $\forall s \exists t (\phi(s, t) \wedge \forall t' (\phi(s, t') \rightarrow t' \doteq t))$ , some authors define it as an axiomatic schema.<sup>†</sup>

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<sup>†</sup>See, e.g., (Enderton 1977; Hrbacek and Jech [1978] 1999; Button 2021).

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# The Axiom of Foundation

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## Axiom of Foundation

$$\forall x (x \neq \emptyset \rightarrow \exists y (y \in x \wedge x \cap y \doteq \emptyset)).$$

# The Axiom of Foundation

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## Axiom of Foundation

$\forall x (x \neq \emptyset \rightarrow \exists y (y \in x \wedge x \cap y \doteq \emptyset)).$

### Category

Axiom determining a property of sets.



# The Axiom of Foundation

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## Axiom of Foundation

$$\forall x (x \neq \emptyset \rightarrow \exists y (y \in x \wedge x \cap y \doteq \emptyset)).^\dagger$$

### Category

Axiom determining a property of sets.

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<sup>†</sup>See the corrections to the textbook.

# The Axiom of Foundation

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## Exercise 4.41(a)

Let  $z$  be a set. Show that  $z \notin z$ .

# The Axiom of Foundation

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## Exercise 4.41(a)

Let  $z$  be a set. Show that  $z \notin z$ .

## Solution

The singleton  $\{z\}$  is a set by the Axiom of Pairs whose unique element is  $z$ . By the Axiom of Foundation

$$\{z\} \cap z \doteq \emptyset.$$

Therefore,  $z \notin z$  (otherwise  $\{z\} \cap z \neq \emptyset$ ).

# The Axiom of Foundation

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## Definition

Let  $\langle x_n \rangle$  be a sequence. An **infinite descending  $\in$ -chain** is a sequence such that  $x_{n+1} \in x_n$ , for all  $n \in \omega$ , that is,

$$\cdots x_{n+1} \in x_n \in \cdots \in x_2 \in x_1 \in x_0.$$

# The Axiom of Foundation

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## Theorem 4.3 (No infinite descending $\in$ -chains)

Let  $\langle x_n \rangle$  be a sequence. The sequence cannot form an infinite descending  $\in$ -chain.

# The Axiom of Foundation

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## Theorem 4.3 (No infinite descending $\in$ -chains)

Let  $\langle x_n \rangle$  be a sequence. The sequence cannot form an infinite descending  $\in$ -chain.

### Proof (by contradiction)

Suppose that  $\langle x_n \rangle$  forms an infinite descending  $\in$ -chain.

Let  $x := \{ x_n : n \in \omega \}$ . By the Axiom of Foundation exists  $y$  such that  $x \cap y = \emptyset$ .

Since  $y \in x$ , then  $y = x_n$  for some  $n$ . Since  $x_{n+1} \in x$  and  $x_{n+1} \in y$  then  $x \cap y \neq \emptyset$ .

Contradiction!

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